



Software Inspections – 3



Lecture # 17

What Inspections Are Not - 1

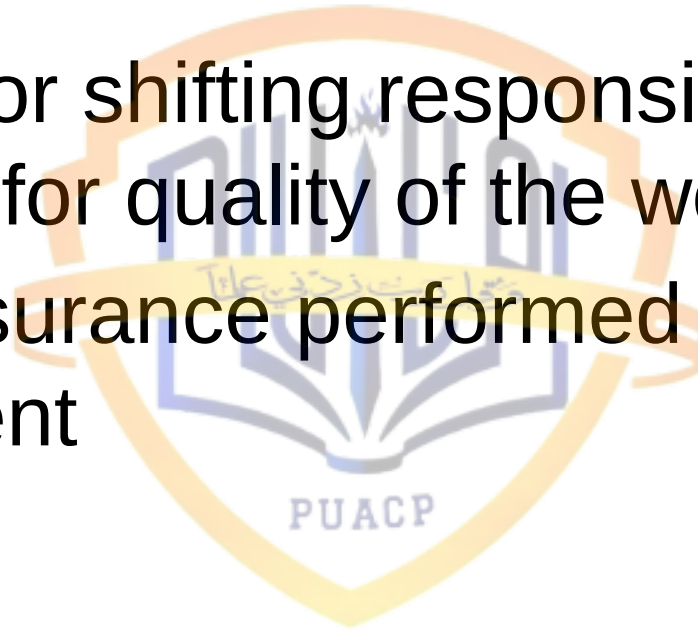
- A review of the style of a work product
- A review of the producer, and especially not a means to evaluate the producer by management
- An impromptu meeting; it is a scheduled meeting with resource considerations to enable effectiveness

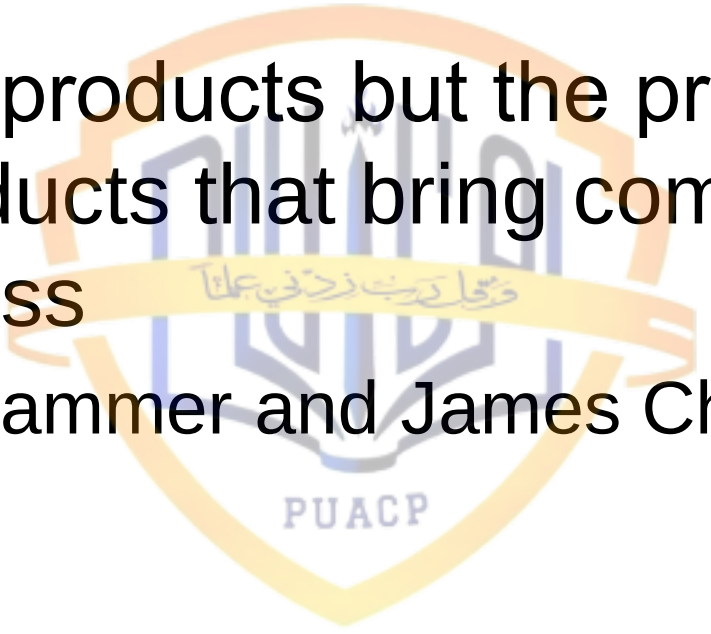
What Inspections Are Not - 2

- A casual or informal meeting; there is structure and rigor for a purpose
- Typically the time or place to fix defects or discuss possible solutions
- Free! But they do yield a high return on investment

What Inspections Are Not - 3

- A vehicle for shifting responsibility to inspectors for quality of the work product
- Quality assurance performed at the end of development

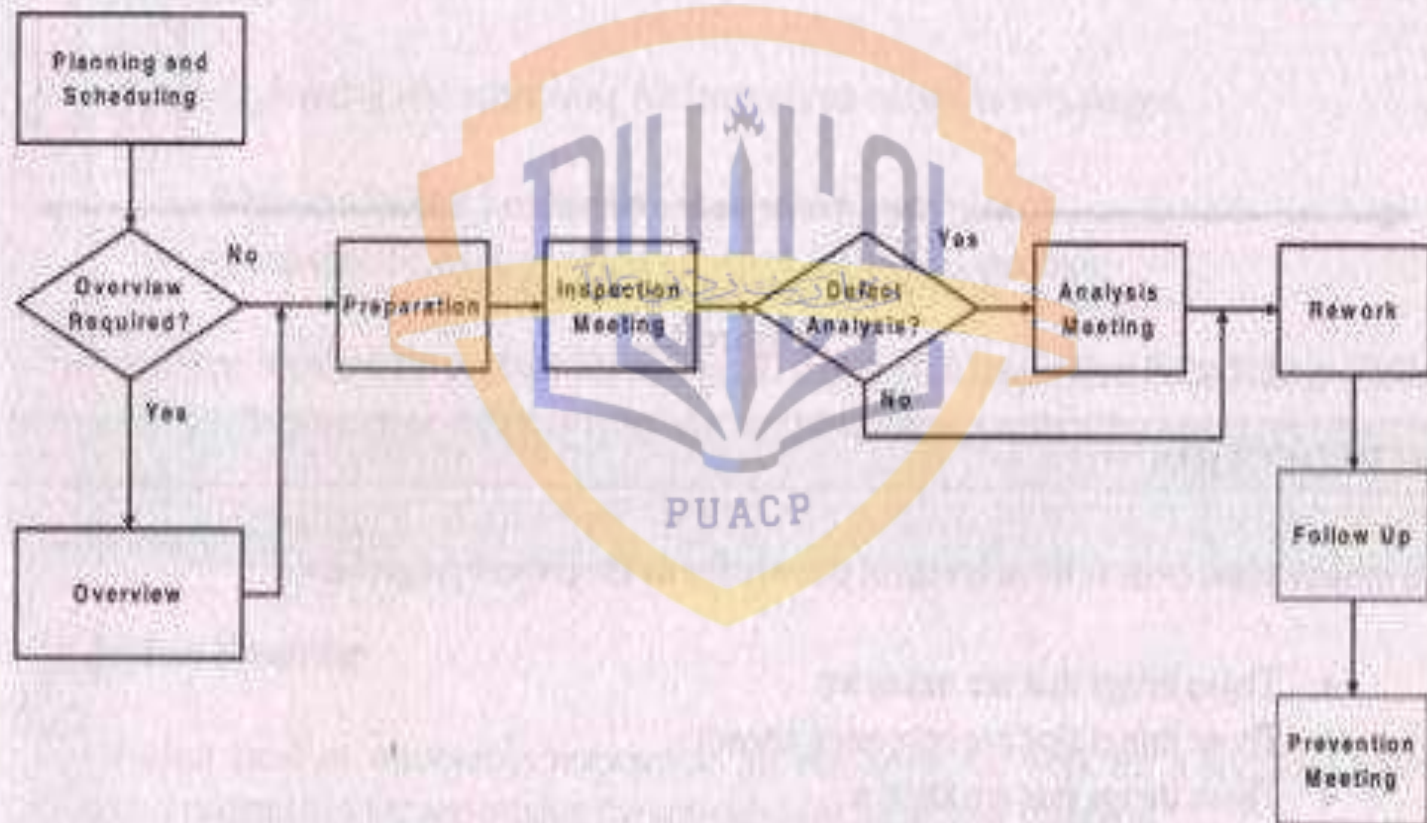


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- It's not the products but the processes that create products that bring companies long-term success
 - Michael Hammer and James Champy



Inspection Process Flow

Inspection Process Flow

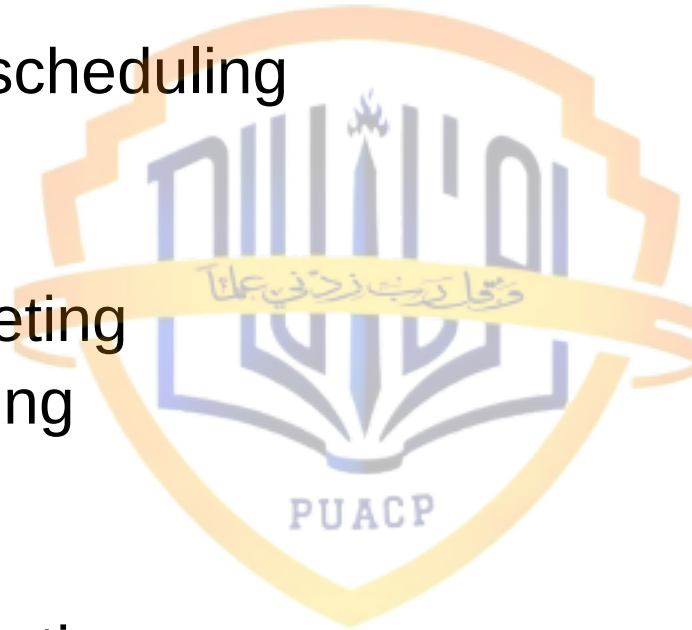




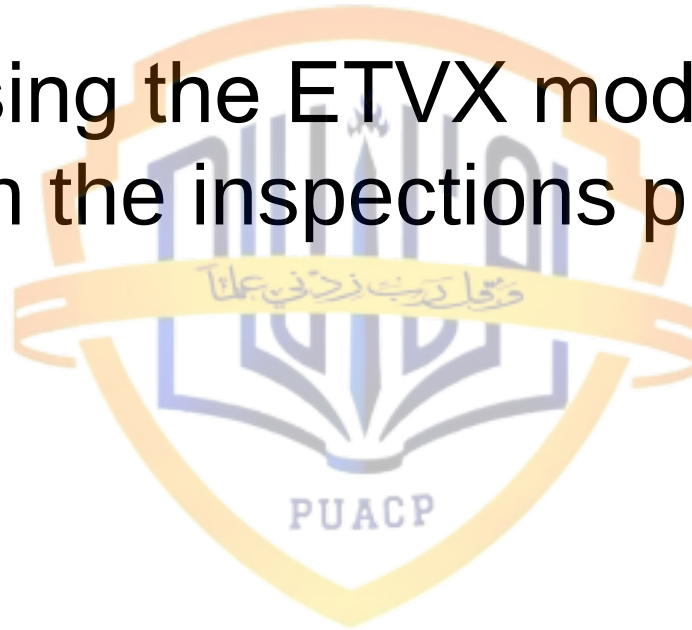
Inspection Process

Inspection Process

- Planning and scheduling
- Overview
- Preparation
- Inspection meeting
- Analysis meeting
- Rework
- Follow-up
- Prevention meeting
- Data recording and reports
- Inspection process monitoring



- We'll be using the ETVX model to describe the steps in the inspections process



Planning and Scheduling - 1

- To ensure adequate time and resources are allocated for inspections and to establish schedules in the project for work products to be inspected, to designate the inspection team, and to ensure the entry criteria are satisfied

Planning and Scheduling - 2

- All project plans exist at three levels of knowledge as the project progresses
 - Those things that are unknown
 - Those things that are becoming known
 - Those things that are known
- Plan details reveal themselves to the planner as a rolling wave

Planning and Scheduling - 3

- The project lead must plan which inspections are to be performed at the initial stages of the project
- Unknowns become knowns
- Has two sections
 - Inspection planning
 - Inspection scheduling

Inspection Planning: Responsibility

- The project lead or whoever is responsible for managing the work for a specified software project is responsible for performing the activities for Inspection Planning

Inspection Planning: Entry Criteria - 1

- A policy exists for inspections in the project's organization
- Planning procedures, including planning for inspections exist
- A project begins and includes the requirement to plan for inspections
- Work product types to be inspected are identified in the project plan

Inspection Planning: Entry Criteria - 2

- Well-defined work product completion or readiness criteria are available
- Initial estimates are provided for the size of the work products to be inspected
- Expected project participants have been trained or a training plan is defined
- Goals and targets have been established for the volume or percentage of work products to be inspected

Inspection Planning: Tasks

- Determine what will be inspected
- Estimate resources for inspections and allocate budget
- Set milestones for the inspections
- Identify dependencies on other groups

Inspection Planning: Validation/Verification - 1

- The SQA group in the organization should assure that the project plan has been documented and includes planned inspections as required by the organization policy and procedures

Inspection Planning: Validation/Verification - 2

- Data to be gathered during this activity
 - Which work products are planned for inspection
 - The estimated size of work products to be inspected
 - Risks
 - The number of planned inspections
 - Planned effort to be spent on inspections

Inspection Planning: Exit Criteria

- There is a project plan showing the inspections to be held, including resources and milestones that may be known in the early stages of the project
- Where milestones may not be known a boundary of probable dates should be noted in the plan for the inspections
- Adequate resources are allocated in the project plan for inspections

Inspection Scheduling: Responsibility

- The project lead is responsible
 - For requesting, selecting, or assigning Moderators when a work product approaches inspection readiness
 - For ensuring the work product will be ready for inspection
 - For ensuring that the participants are made available
 - For making known to a qualified Moderator that an inspection is to be scheduled

Inspection Scheduling: Other Roles

- The moderator with the project lead is responsible for completing inspection scheduling
- This includes
 - Agreement on a specific date
 - Ensuring that entry criteria is met
 - Completing all logistics requirements
 - Scheduling the participants and inspection activities

Inspection Scheduling: Entry Criteria

- A work product is approaching inspection readiness
- Resources are available
- The project lead makes a request to a moderator for an inspection or a set of inspections

Inspection Scheduling: Tasks

- Send a notification that an inspection will be needed
- Determine the inspection meeting date
- Ensure that the work product to be inspected meets entry criteria
- Schedule the inspection meeting

Inspection Scheduling: Validation/Verification - 1

- The moderator remains actively involved during the inspection scheduling period and is responsible for assuring that all tasks up to completion of the inspection meeting are performed
- The SQA groups ensures that a moderator has been assigned

Inspection Scheduling: Validation/Verification - 2

- Data gathered during this activity includes
 - How much in advance the project lead is sending notification to the moderator
 - How long is the period between notification and the inspection meeting
 - How many inspections required postponement
 - Actual versus planned inspections

Inspection Scheduling: Exit Criteria

- The inspection activities have been recorded as Performed on the scheduled dates and Closed within the dates determined at the inspection meeting or rework

Overview - 1

- Provides the inspection participants a background and understanding, when warranted, of the scheduled inspection material
- An overview is not an inspection meeting
- If inspectors are sufficiently familiar with the work product, the overview can be skipped

Overview - 2

- Another reason for an overview meeting, is to identify any open issues in the work product
- An open issue is an acknowledgement of the fact that a subpart of the work product is not complete for some reason
- The producer may want focus the inspectors on subparts that are problematic or of some concern

Overview: Responsibility

- The producer's primary responsibility for the success of the overview meeting is to deliver the presentation
- If overview material is provided, it is the producer's responsibility to make sufficient copies for the meeting either directly or via the moderator

Overview: Other Roles

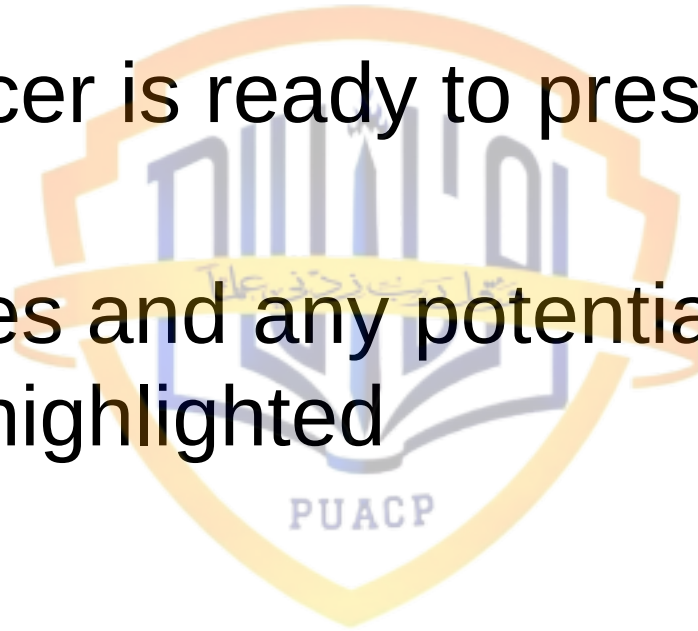
- The moderator determines with the project lead whether an overview is necessary, schedules the overview meeting, obtains the meeting room, and records the time of the meeting, the number of participants, and the results of the meeting
- Inspectors participate during the overview meeting and must concur that the overview met the exit criteria

Overview: Entry Criteria - 1

- A project lead has sent notification for an inspection
- The inspection requires a mandatory overview, or criteria for an optional overview has been satisfied; e.g.,
 - Complexity of the work product solution
 - Volume of material in the work product
 - Criticality of the work product
 - Customer requirements

Overview: Entry Criteria - 2

- The producer is ready to present the overview
- Open issues and any potential problem areas are highlighted



Overview: Tasks - 1

- Producer prepares for the overview using a format and style that will best convey the information to the participants
- Moderator invites the participants to the overview meeting
- Producer presents the overview
- Inspection team members concur that the overview satisfies the needs for preparation and inspection meeting

Overview: Tasks - 2

- Any open issues are documented in the inspection report
- If the overview is used to familiarize the participants with their roles, the inspection process, or some other aspect key to this inspection, the moderator will provide this briefing
- Defects, if any, are documented

Overview: Validation/Verification - 1

- The moderator uses the work product overview meeting entry criteria and procedure to determine if a meeting is necessary
- The inspection team is in concurrence with the decision taken to have an overview or not

Overview: Validation/Verification - 2

- The inspectors have the responsibility to state that the overview, when held, is satisfactory for their preparation and subsequent inspection meeting
- The SQA group ensures that the moderator has used the overview meeting criteria and ensures an appropriate decision was made to have an overview or not. This can be done via audits of the process records or sampling of inspections

Overview: Validation/Verification - 3

- Data gathered during this activity
 - How much participant time was spent in the overview
 - The clock time for the overview
 - Time between notification and the overview meeting
 - How many overviews required rescheduling
 - How many defects were identified at the overview

Overview: Exit Criteria

- The overview meeting was determined to be satisfactory by the inspectors and SQA
- Open issues are documented
- Potential problems areas are noted to the participants for preparation and for the reader for the inspection meeting
- Defects, if any, are documented

Summary

- Discussed planning & scheduling and overview of the software inspections process



References

- Software Inspections by Ronald A. Radice, Tata-McGraw Hill, 2003 (Chapter 3.1 – 3.4)

